

# Desk controller

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## Change history

|       |            |                             |
|-------|------------|-----------------------------|
| 1.0.0 | 23/09/2026 | First release               |
| 1.1.0 | 18/10/2026 | Migration to KiBot template |
| 1.0.0 | TBD        | TBD                         |
| 1.0.0 | TBD        | TBD                         |
| 1.0.0 | TBD        | TBD                         |
| 1.0.0 | TBD        | TBD                         |
| 1.0.0 | TBD        | TBD                         |
| 1.0.0 | TBD        | TBD                         |
| 1.0.0 | TBD        | TBD                         |

System overview



File: System\_overview.kicad\_sch



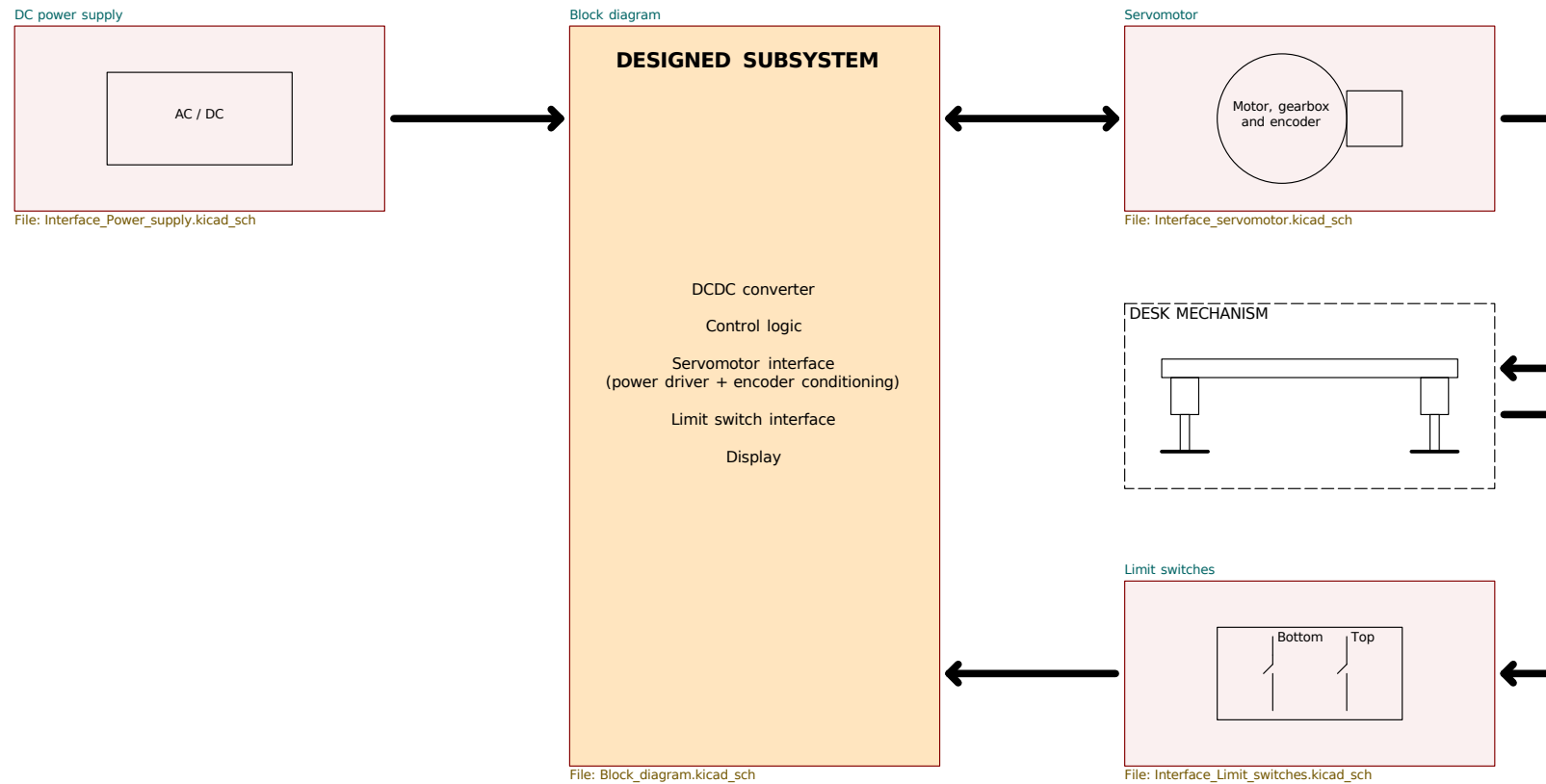
### Desk controller

#### Notes

This is a comment 1 field - potentially not used  
This is a comment 2 field - potentially not used  
This is a comment 3 field - potentially not used  
This is a comment 4 field - potentially not used

|                       |            |
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| Sheet: Root           |            |
| File: ktest.kicad_sch |            |
| Rev: 0.0.1            | Drwn: FLR  |
| Mat: Concept          | Rev'd: FLR |
| Date: 2026-10-06      | RLs'd: FLR |
| Dwg: NA               | Size: A4   |
| Generated with KiCad  | Page: 1/11 |

# System overview



This sheet shows the overall system and the interfaces to external components. External components are shown for system context only and are not part of this schematic. Only the "DESIGNED SUBSYSTEM" is covered by this schematic.

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## Desk controller

### Notes

Sheet: System overview

File: System\_overview.kicad\_sch

Rev:

Mat: Concept

Date:

Dwg: NA

Generated with KiCad

Drwn: FLR

Rev'd: FLR

RLs'd: FLR

Size: A4

Page: 2/11

# DC power supply

## Interface specification

Barrel connector (Dell type, with ID pin)

Pin

- 1 Supply - 19.5 V, 3 A (inner shell)
- 2 ID (internal pin)
- 3 GND (outer shell)
- SH Shield

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### Desk controller

#### Notes

|                                        |            |
|----------------------------------------|------------|
| Sheet: DC power supply                 |            |
| File: Interface_Power_supply.kicad_sch |            |
| Rev:                                   | Drwn: FLR  |
| Mat: Concept                           | Rev'd: FLR |
| Date:                                  | RLs'd: FLR |
| Dwg: NA                                | Size: A4   |
| Generated with KiCad                   | Page: 3/11 |

Interface specification

JST B6B-XH-A / JST XHP-6 connector

| Pin | Description            |
|-----|------------------------|
| 1   | Encoder B output (VCC) |
| 2   | Encoder A output (VCC) |
| 3   | GND                    |
| 4   | VCC (3.3 - 5V)         |
| 5   | Motor B (3 A, 12-24 V) |
| 6   | Motor A (3 A, 12-24 V) |



|                 |                                      |            |
|-----------------|--------------------------------------|------------|
| Desk controller | Sheet: Servomotor                    |            |
|                 | File: Interface_servomotor.kicad_sch |            |
|                 | Rev:                                 | Drwn: FLR  |
|                 | Mat: Concept                         | Rev'd: FLR |
|                 | Date:                                | RLs'd: FLR |
| Notes           | Dwg: NA                              | Size: A4   |
|                 | Generated with KiCad                 | Page: 4/11 |

# Limit switches

## Interface specification

JST B4B XH A / JST XHP-4 connector

### Pin Description

- 1 Bottom limit switch (Normally open, switch closes to GND)
- 2 GND
- 3 GND
- 4 Top limit switch (Normally open, switch closes to GND)

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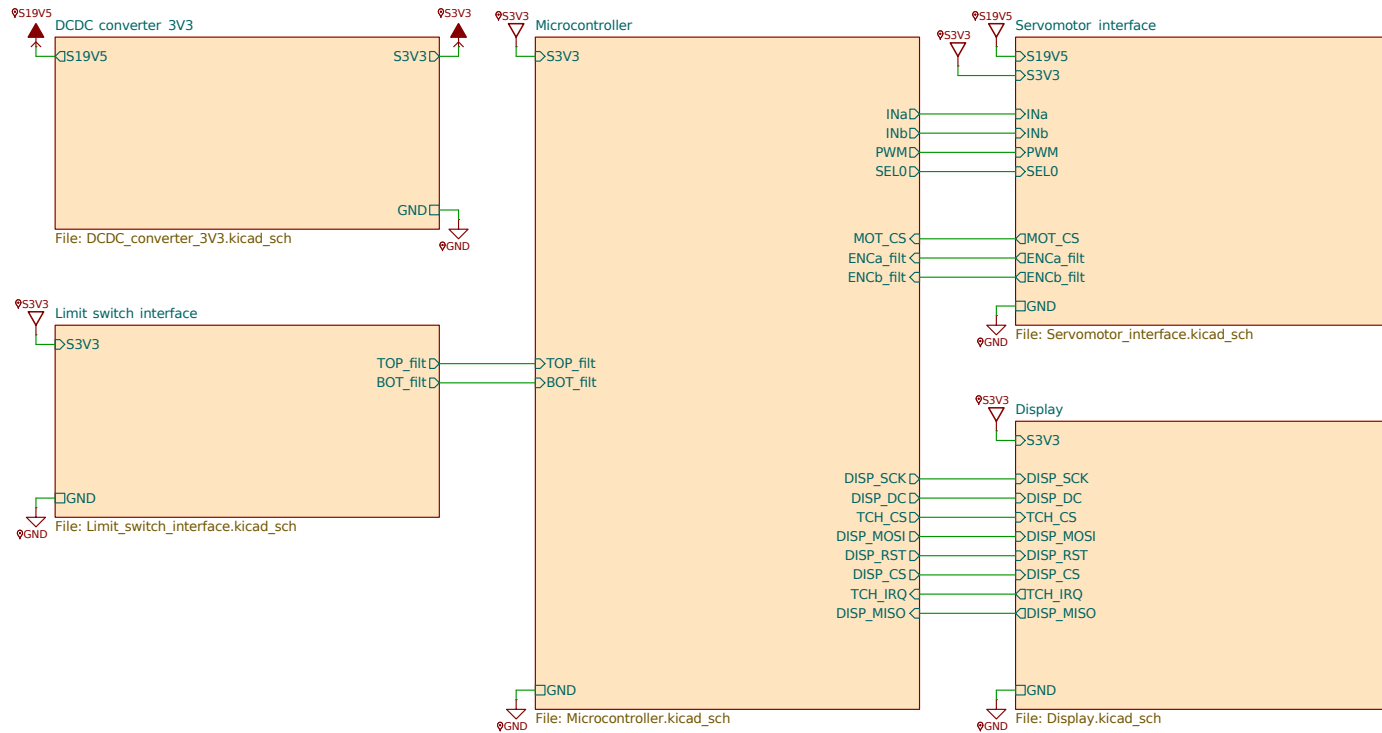


### Desk controller

#### Notes

|                                          |            |
|------------------------------------------|------------|
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| Mat: Concept                             | Rev'd: FLR |
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# Block diagram



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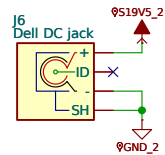
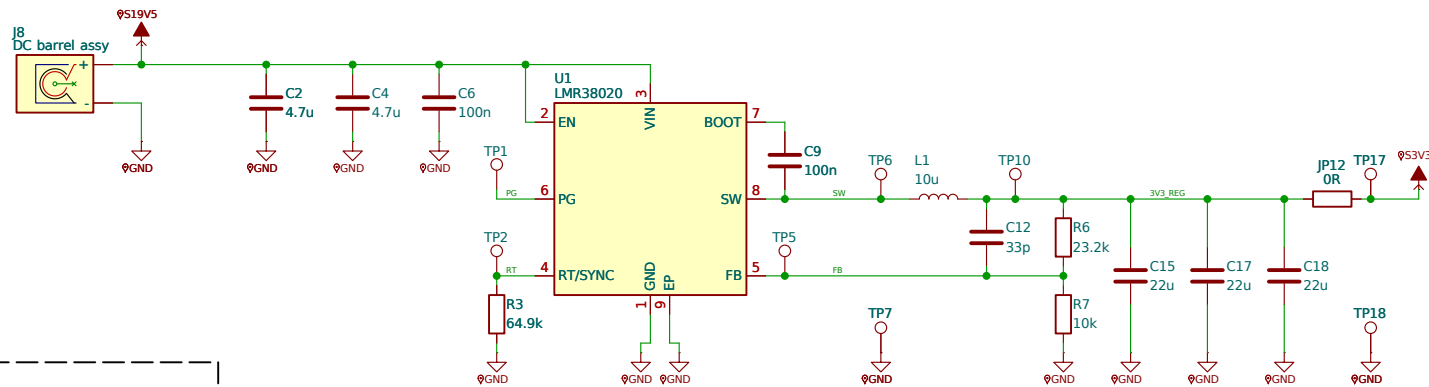
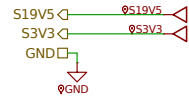
## Desk controller

### Notes

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| Mat: Concept                  | Rev'd: FLR |
| Date:                         | RLs'd: FLR |
| Dwg: NA                       | Size: A4   |
| Generated with KiCad          | Page: 6/11 |

# DCDC converter 3V3

## Block interface



J6 is fitted to a small "standalone" PCB which will then be soldered into the "main" PCB. This standalone PCB + connector are what make the "DC barrel assy" used in J8 position.

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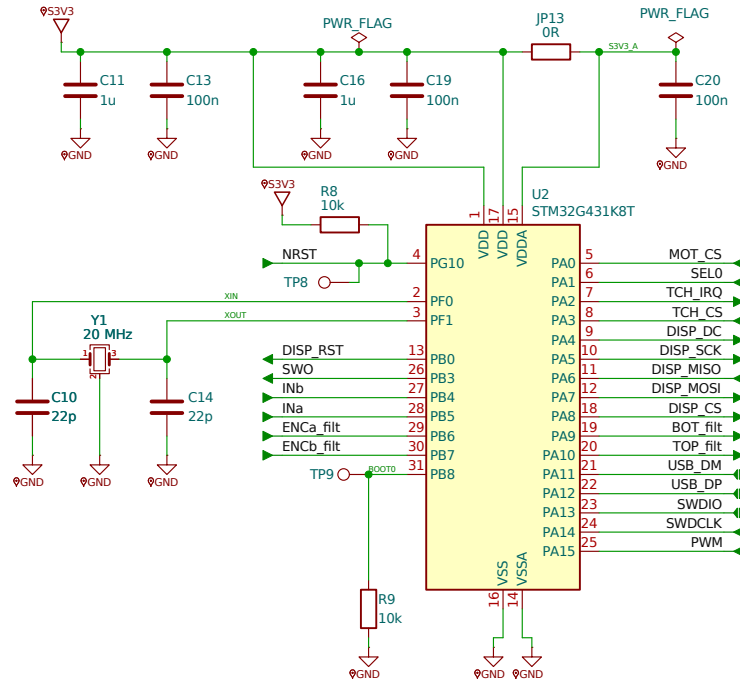
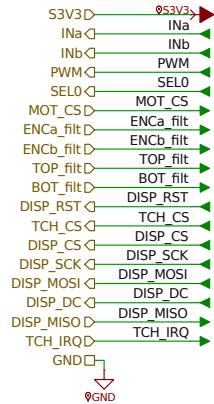


## Desk controller

### Notes

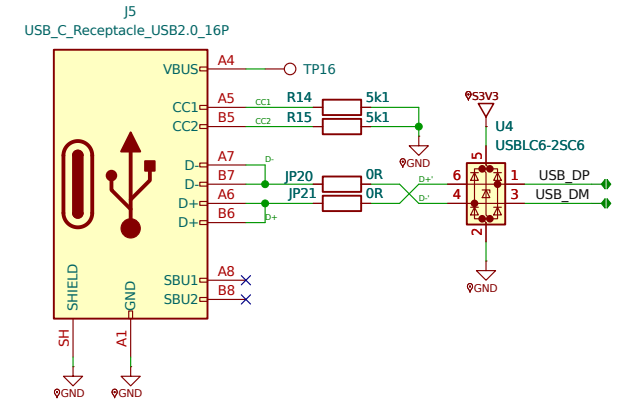
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| Mat: Concept                       | Rev'd: FLR |
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## Block interface

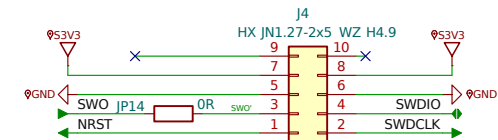


## Microcontroller

### USB interface (serial comms port)



### SWD header



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### Desk controller

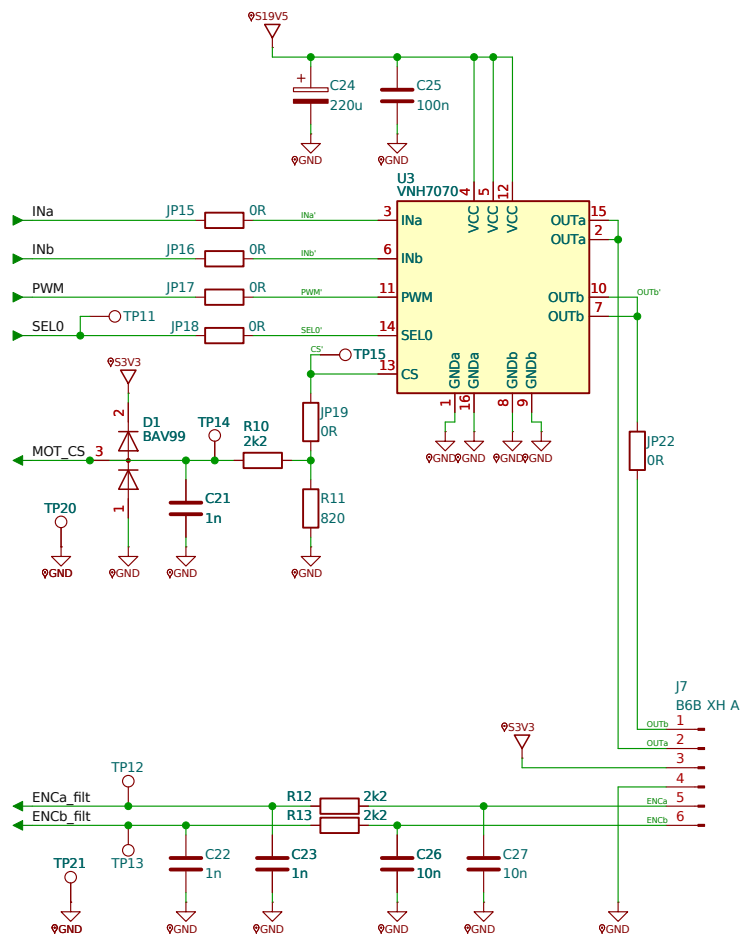
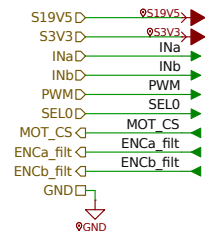
#### Notes

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# Servomotor interface

## Block interface



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Desk controller

Notes

Sheet: Servomotor interface

File: Servomotor\_interface.kicad\_sch

Rev:

Mat: Concept

Date:

Dwg: NA

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Drwn: FLR

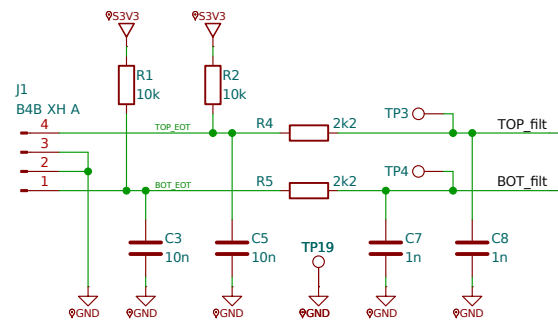
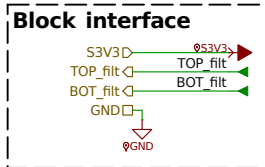
Rev'd: FLR

Rs'd: FLR

Size: A4

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## Limit switch interface

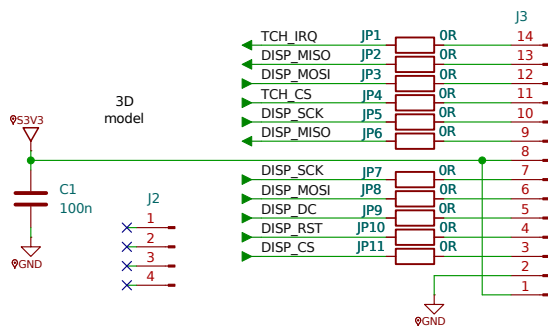
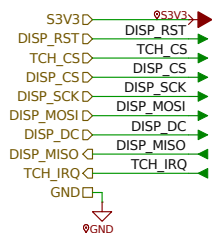


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|                        |                                        |             |
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|                        | Mat: Concept                           | Rev'd: FLR  |
|                        | Date:                                  | RLS'd: FLR  |
| <b>Notes</b>           | Dwg: NA                                | Size: A4    |
|                        | Generated with KiCad                   | Page: 10/11 |

## Block interface



TFT1  
TFT\_2.4in\_320x240



The display will not be directly soldered to the PCB. Instead, generic 2.54 mm pitch sockets J2 and J3 will be used as connectors.

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Desk controller

Notes

Sheet: Display  
File: Display.kicad\_sch

|                      |             |
|----------------------|-------------|
| Rev: Concept         | Drwn: FLR   |
| Date:                | Rev'd: FLR  |
| Dwg: NA              | RLs'd: FLR  |
| Generated with KiCad | Size: A4    |
|                      | Page: 11/11 |